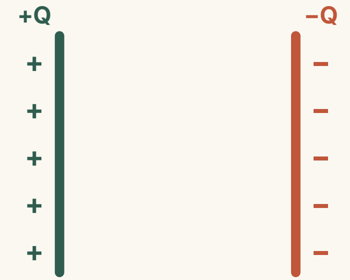
CAPACITANCE

How much charge can one volt hold?

A lesson on capacitors, charge separation, and the meaning of $C = Q/U$.

ESSENTIAL QUESTION

What does a capacitor's capacitance actually tell us?



LESSON MAP

Build the idea in four moves.

01

SEPARATE

How two conductors store equal and opposite charge.

02

MEASURE

Why capacitance means charge stored per volt.

03

DESIGN

How area, spacing, and dielectric change C.

04

APPLY

How to use units, ratings, and $C = Q/U$.

KEEP THIS IN MIND

A capacitor needs two conductors separated by an insulator.

CHARGE STORAGE

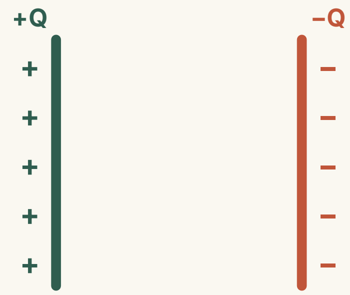
A capacitor stores charge by separating it.

No net charge is created — electrons are moved from one conductor to the other.

1. **Two conductors, one gap**
   Plates A and B separated by an insulator.

2. **Equal and opposite**
   The plates carry +Q and −Q of equal magnitude.

3. **Separation, not creation**
   Electrons simply shift from one plate to the other.



SAME PRINCIPLE

From Leyden jars to modern components, the structure is unchanged.

Two conductors and an insulator — the design has stayed the same for centuries.

EARLY CHARGE STORAGE

The Leyden jar: metal foil inside and out, glass between — a capacitor in all but name.

MODERN CAPACITORS

Rolled films or stacked layers — still just conductors plus an insulator.

A battery moves electrons until the voltage matches the supply.

More separation → larger potential difference, until the capacitor voltage equals the source.

DEFINITION

Capacitance is charge stored per volt.

Q is the charge on either plate; U is the potential difference between them.

$$C = \frac{Q}{U}$$

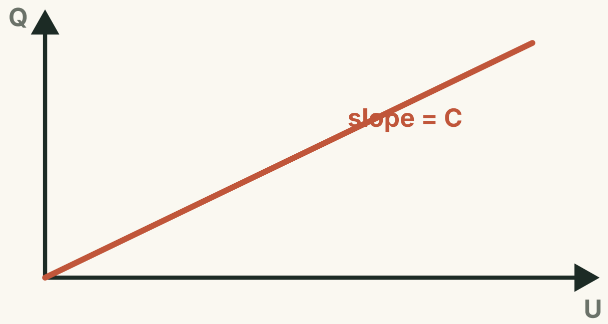
AT THE SAME VOLTAGE

A higher-capacitance device stores a larger charge Q.

Q–U RELATIONSHIP

For one capacitor, Q ∝ U — and the slope is C.

Changing Q and U moves the operating point along the same straight line.



The slope stays fixed because C is a **property of the capacitor**, set by its build.

KEY DISTINCTION

C does not increase just because Q increases.

ANALOGY

Water storage helps — until the analogy misleads.

Amount of water ↔ separated charge Q; container capacity ↔ capacitance C.

USE IT FOR COMPARISON

At the same “pressure”, a larger-capacitance device holds more charge.

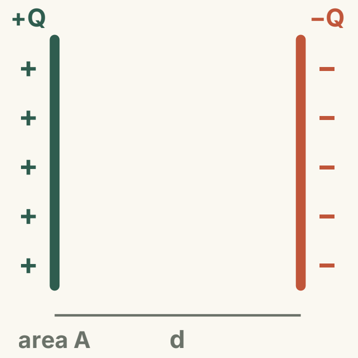
BUT REMEMBER

Voltage is potential difference, not fill height. Charge is separated across two conductors.

PHYSICAL DESIGN

Geometry and dielectric set capacitance.

Three build choices control how much charge the plates can hold per volt.



PARALLEL-PLATE CAPACITOR

$$C = \frac{\varepsilon A}{d}$$

Larger area A → larger C; smaller gap d → larger C; larger ε → larger C.

CONTROLLED COMPARISONS

Three design changes make capacitance larger.

Each one lets the plates separate more charge at the same voltage.

01 · More area

A ↑ — more charge can face across the gap.

02 · Smaller gap

d ↓ — opposite charges interact more strongly.

03 · Dielectric

ε ↑ — polarization reduces the voltage per Q.

UNITS & SCALE

One farad is large, so most components use sub-units.

Capacitance spans many orders of magnitude.

F · **farad**	**mF** · **millifarad**	**µF** · **microfarad**	**nF** · **nanofarad**	**pF** · **picofarad**
1 F	10^{-3} F	10^{-6} F	10^{-9} F	10^{-12} F

DEFINITION $1\,\mathrm{F} = 1\,\mathrm{C/V}$

VOLTAGE LIMITS

Capacitance is useful only below the voltage limit.

Stay at or below the rated voltage; beyond breakdown, the dielectric conducts.

NORMAL OPERATION

From 0 V up to the rated voltage, the dielectric insulates as intended.

BREAKDOWN

Above the breakdown voltage the dielectric conducts — and the part fails.

WORKED EXAMPLE

Use C = Q/U by choosing the unknown first.

A 220 μF capacitor is connected to 12 V. What charge is stored on either plate?

1. **Rearrange**

   $$Q = CU$$

2. **Convert**

   $$220\,\mu\text{F} = 220 \times 10^{-6}\ \text{F}$$

3. **Calculate**

   $$Q = (220 \times 10^{-6})(12) = 2.64 \times 10^{-3}\ \text{C}$$

ANSWER

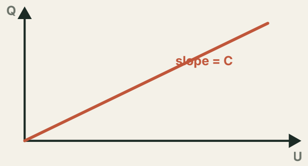
2.64 mC on each plate — equal magnitude, opposite sign.

SUMMARY

Capacitance is charge stored per volt — a property of the build, not the charge.

THE DEFINING RELATION

$$C = \frac{Q}{U}$$



What's stored: equal, opposite charges across two conductors.

What sets C: larger area A, smaller gap d, larger dielectric ε.